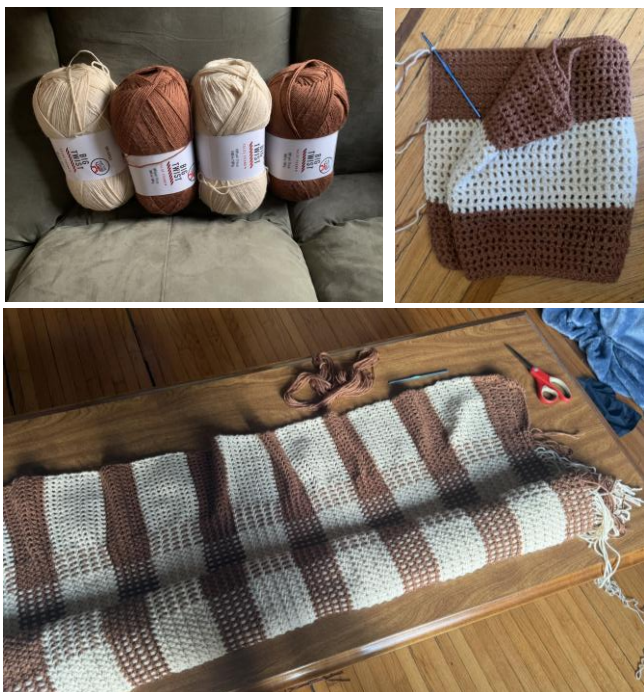




EXPLORING MATHEMATICAL CONCEPTS THROUGH CROCHET

Honors Contract – Spring 2025

Over 95 hours of learning, mistakes, and mathematics compacted into a handmade, crochet blanket.

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Overview

Goal

With my basic knitting and sewing skills, I took on the challenge of learning to crochet. For my final creation, I set out to make a small blanket. I initially followed a basic pattern and then adapted to the challenges that came along the way. Alongside the creation of a blanket, I kept a time log with row counts and calculated averages to be able to explore the mathematics of crochet even further.

Challenges

This project involved a variety of challenges. The first challenge came at being completely new to the art. So, not wanting to jump off the deep end right away, I took the time to create three granny squares and a winter hat before getting started on my final project of a small blanket. With these initial projects, I was able to practice basic stitches and learn that the yarn I initially selected was difficult to work with and not a good weight for my blanket. While creating my final blanket I ran into many challenges as well.

Despite my best efforts to align my edges and keep track of rows, I ended up skipping and then re-adding an additional two rows in the beginning stages of my blanket. These mistakes are less noticeable when looking at the pre-woven blanket. However, weaving in the additional rows required some improvisation.

Relation to Mathematics

Math is an integral aspect of crochet from counting stitches to following basic and complex patterns. The patterns themselves are physical sequences that visualize geometry. And patterns that increase and decrease stitches can lead to visualizing three-dimensional shapes. As I am new to the art, I did not want to take on this challenge right away. However, I did find an interest in the math around crochet that is not necessarily seen within the project itself. So, I will be exploring the math inside and outside of my project further.

Relationship Between Crochet & Mathematics

Mathematics in Crochet & Pattern Breakdown

Stitch counts are one of the most basic, yet most important parts of a crochet project as a consistent stitch count is vital to ensuring straight edges. Along with stitch count, understanding the stretch of a stitch is necessary. This ensures that the stitch count is accurate to maintain the shape of the project. For instance, in my final blanket, I used

double crochet stitches where their height can stretch to three-times their width. So, when chaining my first row, I added an additional three chains to the desired length so that the double crochet would make a proper square shape. Chaining to this desired length also involves some calculations as laid out below.

$10 \text{ double crochets per row} + 10 \text{ chains per row} = 20 \text{ chains needed for every row}$

$20 \text{ chains per row} * 11 \text{ rows of 10 weavings} = 220 \text{ chains for length}$

$220 \text{ chains for length} + 3 \text{ chains for double crochet shape} = 223 \text{ chains}$

The second row of the blanket works with this chained row to start building the structure. Each double crochet is followed by a chain stitch and each double crochet is one stitch apart from the previous. This process is repeated for the rest of the row, each ending with a chain before moving to the next row.

Once the first ten rows are completed, switching colors is done by looping over the next color half-way through the last double crochet. This is shown in the photo to the right. Two colors of yarn are alternated every ten rows for a total of 140 rows or 7 sections of each color. Once this process has been completed, the base blanket is done.



Then, the process of weaving begins. A progress photo of this stage is shown to the left. For weaving, on a flat surface, two strands of yarn are weaved through each row created by the gaps of the chains. The yarn easily twists out of place, so it is necessary to go slow with the darning needle. Once each row is weaved through, the two strands of yarn are looped through and tied together to keep the strands in place. These strings are cut to be the same size once all of the rows have been completed.

Steaming

There are many ways to soften, stretch, and overall block acrylic yarn pieces. For this project, I used steam blocking. This enabled me to stretch my blanket slightly while softening the yarn. This complicates math in crochet because the yarn itself is not entirely predictable for a new crafter. The yarn may stretch, or not, to unexpected lengths that makes the actual dimensions of the project difficult to control.

Lessons for the Future

If I were to retry this pattern, I would ensure to be more careful and cautious of messing up the base mathematics of the piece. Most of my mistakes were made because I did not want to constantly count my stitches. However, I now know that the shape of the blanket is reliant on this count.

Time Spent

Project Breakdown

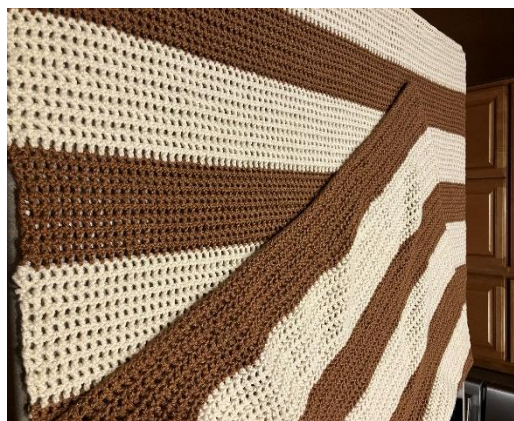
The first three days of my project I was able to smoothly start and complete the first 24 rows of my project. With small distractions and going back to fix a few missed stitches, I was able to avoid making any large mistakes until eleven days into the base blanket. This is when I learned a new term to ‘frog’ as in to undo one or multiple stitches to fix a mistake, repurpose the yarn, etc. On this day, I added two double crochets into the same previous double crochet and had to frog an entire row in order to go back and fix it. For this mistake, I ended



up having to spend extra time and lowered my rows per hour to almost day two levels (1.4 rows/hour). This mistake is pictured to the left and I actually made the same mistake two more times without realizing. This led to difficulties during the weaving process. I continued to add rows, switching back and forth between colors every ten counts. During this process, I went on vacation for Spring Break and brought my new hobby to New Orleans and then on a cruise in the Gulf of Mexico.

This occurred right before my largest mistake (pictured to the right). In this, I had to frog and redo two rows because I stopped and moved to the next row far before the end of the previous row. This mistake also lowered my row per hour count to nearing day two levels (1.4 rows/hour). Despite this





frustration, I kept working and reached 60 hours on the 31st of March. Five days later I finished the base blanket (shown to the left). I then spent an additional 12 days weaving the rows through the blanket. During this process, I had to make sacrifices on perfect row counts to ensure that rows I had dropped half-way through the blanket were still weaved through. Finally, I pinned and steamed the blanket to soften and stretch the yarn for a more complete look.

Project Log – Mathematical breakdown

From March 1st, 2025 to April 19th, 2025, I spent a total of 95.52 hours crocheting, weaving, and finally steaming my blanket. I was able to complete 11 rows in 7.5 hours on the first day. The next two days I completed a combined 13 rows in 8.5 hours. So, for the first three days of crocheting the base blanket, I completed an average of 1.63 rows an hour. In comparison, the last three days of crocheting the base blanket took an average of 2.43 rows an hour. Figure 1 below shows the average rows/hour as I began to improve my skills. I've used 'Day Count' to account for me not crocheting every day. It is also interesting to note that most of my consistent, smaller mistakes were made within the first ten days.

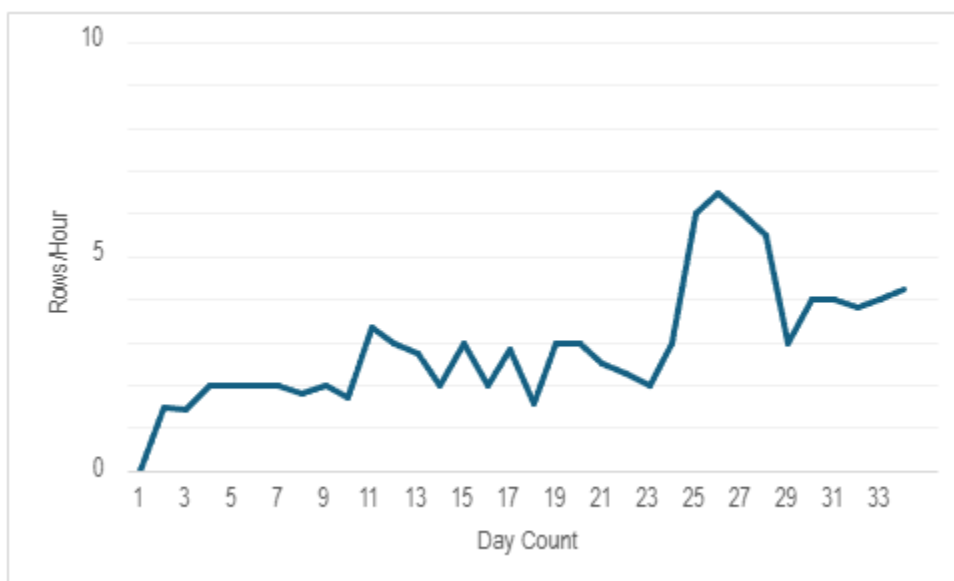


Figure 1: Rows/Hour Over Days Worked. This graph does not include the final day of the project as it took an hour to steam the blanket and every row was affected. This graph also does not include the creation of this paper.

To further dive into Figure 1, a few important days should be highlighted. For instance, days 11 through 15 occurred while on vacation and days 17 and 18 involved fixing a larger mistake, having to frog two rows, and working through a giant knot in the yarn skein. Then, on day 23, I finished the base blanket and rows/hour began greatly increasing as I switched to weaving the yarn. The decrease shown in the days continuing from day 25 can be accounted for by having to work around mistakes made in the base blanket as well as the process becoming tedious. I found the repetitiveness of the crocheting to be relaxing and the repetitiveness of the weaving to be backbreaking as the only flat surface I had available was a coffee table.

As an overall, for the base blanket, I completed an overall average of 2.29 rows per hour. The process of weaving the blanket was 2.05 times faster with an overall average of 4.71 rows per hour. Due to the additional hour of steaming time and the two ‘mistake rows’ resulting in a .5 count each, I completed a total of 253 rows in 95.52 hours for a complete, overall average of 3.02 rows per hour.

Final Results

Final Photos



Inspired Application

During the completion of this project, I was inspired to create a KMeans clustering program for my machine learning course. The program allows users to input a photograph and receive yarn suggestions based on the dominant colors of the photo. This unsupervised learning algorithm expands far beyond the mathematical applications I used to create my blanket, but it highlights how important and valuable this crossover of crochet and mathematics can be.

An example of the input and results of this application are below.



```
Color 1: RGB [174.80044053 163.39368576 145.14287812] -> Red-Heart - Buff (#b2a28c)
Color 2: RGB [56.85873036 31.0261847 12.69804573] -> Caron - Espresso (#311a14)
Color 3: RGB [224.17927701 213.60214376 193.36002522] -> Caron - Off White (#e0d4c9)
Color 4: RGB [45.94302871 53.96001757 99.42735794] -> Bernat - Denim Heather (#393a5d)
Color 5: RGB [125.75883576 108.81860707 95.16969854] -> Red-Heart - Caf   Latte (#6d5f52)
Color 6: RGB [12.53409892 17.32205097 39.59705853] -> Bernat - Navy (#1c1c28)
```

The program is more thoroughly described in the GitHub project found here:

<https://github.com/Emily-McNett/Machine-Learning-Portfolio/tree/main/KMeans-Clustering>.